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# Improving Emotional Text-to-Speech in Low-Resource Languages via Reinforcement Learning

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Hao-Chun Hsieh Pao-Hua Hsu Nien-Tse Wu

Institute of Artificial Intelligence Innovation

National Yang Ming Chiao Tung University

{handsome.ii14, paohuah.ii14, mattwu.ii14}@nycu.edu.tw

## 1 Project Overview

### 1.1 Profile

#### Track: 3. Application

**Abstract and project goal:** We address the problem of emotional capability loss when an expressive TTS model trained on a high-resource language is adapted to a low-resource language. Although the original model can generate emotionally expressive speech, its emotional controllability and prosodic richness often degrade after fine-tuning on limited target-language data. To tackle this issue, we propose an RL-enhanced adaptation framework built upon IndexTTS2 to preserve and realign emotional expressiveness during low-resource language transfer [Zhou et al., 2025]. Our method aims not to learn emotion from scratch, but to retain the emotional prior of the pretrained expressive model while improving emotion intensity, emphasis control, and speech naturalness in the low-resource language. In our study, we instantiate this setting using Chinese as the high-resource source language and Taiwanese as the low-resource target language.

**TL;DR:** This project studies how to preserve and recover the emotional expressiveness of an expressive TTS model when adapting it from a high-resource language to a low-resource language, using reinforcement learning to improve emotion retention and prosodic control.

### 1.2 Motivation

Expressive speech synthesis is increasingly important for applications such as dubbing, broadcasting, and interactive media. However, when a strong expressive TTS model trained on a high-resource language is adapted to a low-resource language, the model often retains intelligibility but loses emotional expressiveness and fine-grained prosodic control. This makes low-resource adaptation not only a language transfer problem, but also an emotion preservation and alignment problem.

- **Why is the problem interesting?** This problem is interesting because it lies at the intersection of low-resource language adaptation, cross-lingual transfer, and controllable expressive speech synthesis. Rather than learning emotional speech generation from scratch, we focus on preserving and realigning the expressive capability that already exists in the pretrained TTS model when transferring it to a low-resource language.
- **Challenge 1: Emotional Forgetting During Low-Resource Adaptation.** When an emotionally expressive TTS model trained on a high-resource language is fine-tuned on limited low-resource language speech data, it often suffers from catastrophic forgetting, causing the synthesized speech to become flatter and less expressive.
- **Challenge 2: Loss of Fine-Grained Prosodic Control.** After low-resource adaptation, the model may still produce intelligible speech, but fail to preserve natural rhythm, pause structure, emphasis, and emotional intensity that are essential for expressive speech generation.

- **Challenge 3: Difficulty of Cross-Lingual Emotion Alignment.** Emotional cues and prosodic patterns do not transfer perfectly across languages. It is therefore difficult to preserve the expressive prior learned from a high-resource language and align it with the pronunciation, prosody, and linguistic structure of a low-resource language using only limited target-language data.
- **Why the problem remains open or unsolved?** This problem remains open because low-resource languages lack sufficient emotionally annotated speech data for conventional supervised training. As a result, naive fine-tuning often prioritizes pronunciation adaptation over expressive preservation. While seminal foundational models like F5-TTS [Chen et al., 2025] achieve high fidelity, they still lack fine-grained emotional controllability. Conversely, works like EmoSphere++ [Cho et al., 2024] demonstrate strong emotion control but are primarily designed for high-resource scenarios. Extending these capabilities to low-resource languages without catastrophic forgetting remains highly challenging. Several prominent recent works have attempted to enhance expressiveness and language adaptability, including the IndexTTS2 [Zhou et al., 2025] and IndexTTS 2.5 [Li et al., 2026] systems, CosyVoice 3 [Du et al., 2025], and EMORL [Li et al., 2025]. Despite these advances, maintaining deep emotional controllability and prosodic richness during low-resource adaptation without losing the pretrained expressive prior is still an unsolved problem.
- **State-of-the-Art (SOTA) Method:** Currently, IndexTTS2 provides a strong foundation for zero-shot and controllable speech synthesis [Zhou et al., 2025]. In this project, we use it as the expressive base model and investigate how reinforcement learning can help preserve and realign its emotional capabilities during adaptation from a high-resource language to a low-resource language.

## 2 Problem Formulation

We formulate emotional preservation and prosodic realignment during low-resource language adaptation as a fully observable Markov Decision Process (MDP). Operating within an autoregressive Text-to-Semantic (T2S) environment, the RL agent (a language model policy  $\pi_\theta$ ) observes a state  $s$  encompassing available conditioning information: an input text sequence, a speaker timbre embedding, a duration constraint embedding, and a target global emotion embedding. The action  $a$  is the generation of a discrete semantic token sequence, which is subsequently converted into an acoustic waveform. Upon generating a complete sequence, the RL learner receives a multi-objective scalar reward signal  $R(s, a)$  computed from the decoded speech. This composite feedback vector, formalized as  $R = \lambda_1 R_{\text{emo-clf}} + \lambda_2 R_{\text{intensity}} + \lambda_3 R_{\text{emphasis}} + \lambda_4 R_{\text{ASR}}$ , balances four task-specific scalar components: categorical emotion classification accuracy, global emotion intensity (distance to a neutral centroid in Valence-Arousal-Dominance space), local emphasis control (relative pitch and energy patterns), and an ASR-based content reward for semantic intelligibility.

The optimization problem seeks to update the policy parameters  $\theta$  by maximizing the expected multi-objective reward using Group Relative Policy Optimization (GRPO). The maximization objective,  $\mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} [R(s, a) \nabla_\theta \log \pi_\theta(a|s)]$ , is subject to a Kullback-Leibler (KL) divergence penalty constraint,  $\beta \text{KL}(\pi_\theta(\cdot|s) \| p_{\text{adapt}}(\cdot|s))$ . This constraint anchors the policy to a base model  $p_{\text{adapt}}$  to prevent catastrophic deviation and preserve the pretrained high-resource expressive prior. Our formulation rests on two key assumptions: (1) an initial parameter-efficient adaptation stage successfully aligns the model to the target low-resource language while keeping most expressive knowledge intact, and (2) the discrete semantic token space produced by the T2S module is sufficiently expressive to retain and recover subtle emotional intensities and local prosody variations during final acoustic reconstruction.

## 3 Empirical Evaluation

To rigorously evaluate the effectiveness of our proposed RL-enhanced IndexTTS2 framework in preserving and recovering emotional expressiveness during transfer from a high-resource language to a low-resource language, we employ a combination of objective and subjective evaluations.

### 3.1 Performance Metrics

We plan to adopt several key performance metrics to comprehensively assess speech intelligibility, speaker fidelity, emotional expressiveness, and emotion retention after low-resource adaptation.

- **Objective Metrics:**
  - **Word Error Rate (WER):** Evaluates speech intelligibility and articulation accuracy using ASR models like FunASR [An et al., 2025] or Whisper [Radford et al., 2022].
  - **Speaker Similarity (SS):** Computed as the cosine similarity between speaker embeddings extracted from a pretrained speaker recognition model to ensure the timbre is maintained after low-resource adaptation and RL optimization.
  - **Speech Emotion Recognition (SER):** Measured using the open-source emotion2vec model to objectively verify if the generated speech matches the target emotion category.
- **Subjective Metrics (MOS Framework):**
  - **Multidimensional MOS:** We will conduct human evaluations based on a 1-5 scale for Similarity MOS ( $\uparrow$ ), Intelligibility MOS ( $\uparrow$ ), and Naturalness MOS ( $\uparrow$ ).

### 3.2 Baseline Methods

We compare our method against several baselines to explicitly quantify emotional forgetting and emotional recovery during low-resource adaptation.

- **CosyVoice2** [Du et al., 2024]: A strong text-to-speech baseline fine-tuned on the target low-resource language for cross-model comparison.
- **CosyVoice3** [Du et al., 2025]: An improved model built upon CosyVoice2, which is also fine-tuned on the target data to evaluate cross-lingual generalization and speech generation quality.
- **IndexTTS2 + Naive Low-Resource Fine-Tuning:** A baseline that fine-tunes IndexTTS2 on limited target-language data without any explicit preservation mechanism, used to measure the severity of emotional forgetting.
- **IndexTTS2 + Parameter-Efficient Adaptation:** A target-language adaptation baseline using a lightweight adaptation strategy, used to test whether preserving more pretrained parameters helps retain emotional expressiveness.
- **IndexTTS2 + Parameter-Efficient Adaptation + RL (Ours):** Our full method, which applies RL-based emotional and prosodic alignment after low-resource adaptation.

### 3.3 Benchmark Datasets

To train the low-resource adaptation model and evaluate catastrophic forgetting of emotional expressiveness, we employ the following datasets:

- **Emotion Supervision Data:** We utilize the Emotional Speech Database (ESD) and the Espresso dataset, which provide annotations for discrete emotions, continuous intensities, and emphasis. These serve as supervision signals for RL-based emotional realignment training.
- **Taiwanese Across Taiwan Corpus (TAT)** [Liao, 2022]: A NAER-hosted large-scale native Taiwanese speech corpus. We utilize the TAT-MOE subset, containing approximately 200 hours of recordings, as the training set for low-resource fine-tuning. For validation and testing, we use the evaluation and test sets from TAT Volume 1 (TAT-Vol1), each comprising approximately 5 hours of data.

## 4 Methodology

Our methodology follows a two-stage framework built upon a pre-trained expressive IndexTTS2 model. Starting from a model trained on a high-resource language, we first adapt it to the low-resource target language, and then apply reinforcement learning to preserve and recover emotional expressiveness, prosodic control, and content accuracy during transfer.

### 4.1 Stage 1: Low-Resource Language Adaptation

In the first stage, we adapt the expressive base model to the low-resource target language. The goal of this stage is to acquire target-language pronunciation and linguistic patterns while preserving as

much of the pretrained expressive prior as possible. To reduce severe catastrophic forgetting, we adopt a parameter-efficient adaptation strategy, such as lightweight adapter modules or LoRA, so that only a limited subset of model parameters is updated during target-language transfer.

## 4.2 Stage 2: RL-Based Emotional Realignment

In the second stage, we introduce an RL-based alignment procedure [Li et al., 2025] on top of the adapted model. The core idea is to treat the autoregressive generation of semantic speech representations as a sequential decision-making process, while computing task-specific rewards from the decoded speech waveform. This stage explicitly optimizes emotion preservation, emotional intensity, local emphasis, and content intelligibility in the low-resource language. In practice, the RL updates are mainly applied to the autoregressive T2S transformer, and can optionally include the emotion- or style-related conditioning modules when stronger expressive realignment is needed.

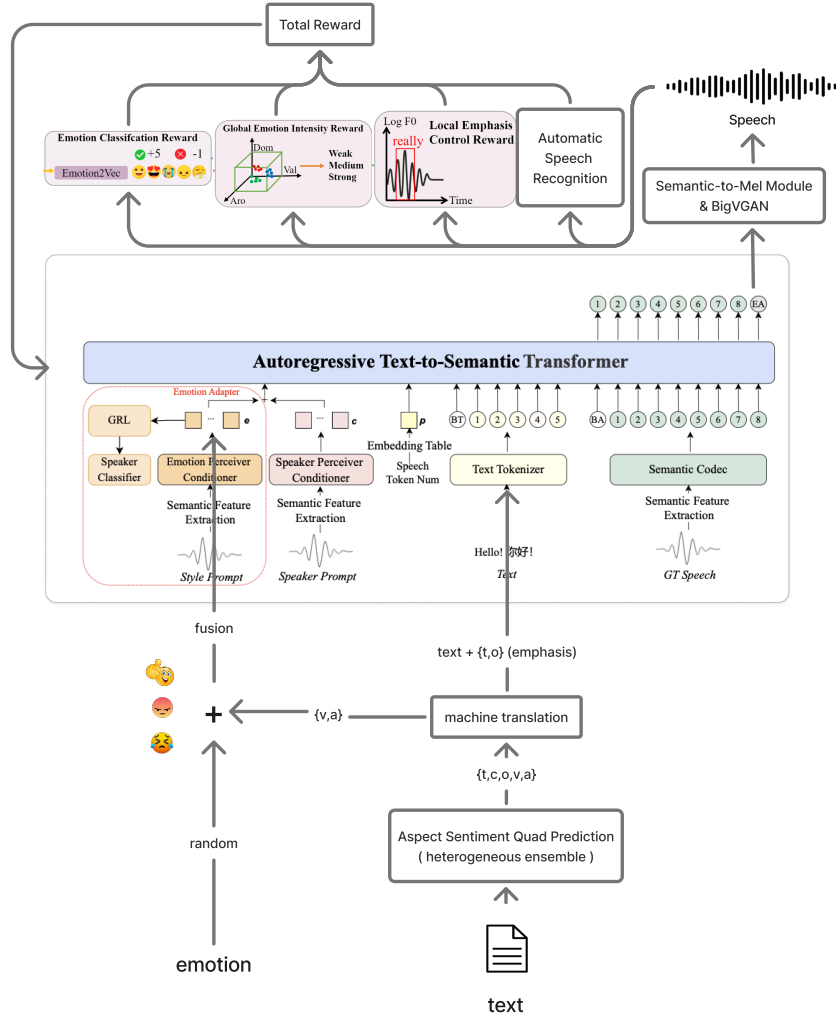


Figure 1: High-level architecture of the proposed RL-enhanced IndexTTS2 framework. After low-resource adaptation, the generator outputs semantic speech tokens, which are decoded into speech waveforms and evaluated by emotion, intensity, prosody, and content reward models. These rewards are used to update the policy via Group Relative Policy Optimization (GRPO), mitigating emotional forgetting during language transfer.

### 4.3 Reinforcement Learning Formulation

The generation of semantic speech tokens is formulated as a Markov Decision Process (MDP), while the reward is computed from the decoded speech waveform:

- **State ( $S_t$ ):** The hidden states of the IndexTTS2 autoregressive decoder, incorporating the input text, speaker conditions, and previously generated speech tokens up to step  $t$ .
- **Action ( $A_t$ ):** The predicted semantic speech token at step  $t$ , sampled from the policy distribution  $\pi_\theta(A_t|S_t)$ .
- **Reward ( $R_t$ ):** A composite reward signal provided upon the completion of a sequence or segment. The reward is designed to preserve emotional expressiveness while maintaining intelligibility in the low-resource target language, and comprises four components:
  1. *Emotion Classification Reward:* Evaluated using a pre-trained emotion recognition model (e.g., emotion2vec) to measure whether the generated speech matches the target emotion category.
  2. *Global Emotion Intensity Reward:* Computed from the distance between the generated speech and a neutral centroid in the Valence-Arousal-Dominance (VAD) space, in order to encourage the target global emotional intensity.
  3. *Local Emphasis Control Reward:* A reward based on the relative pitch and energy patterns of emphasized words, which encourages correct placement of local stress and expressive prosody.
  4. *ASR-Based Content Reward:* Computed using an ASR model on the generated speech to compare the transcription against the input text, encouraging content preservation and intelligibility during RL optimization.

In summary, the sequence-level reward can be written as

$$R = \lambda_1 R_{\text{emo-cl}} + \lambda_2 R_{\text{intensity}} + \lambda_3 R_{\text{emphasis}} + \lambda_4 R_{\text{ASR}},$$

where the coefficients  $\lambda_i$  control the trade-off between emotional expressiveness, local prosody, and content accuracy.

### 4.4 Alleviating Catastrophic Forgetting

To address the degradation of emotional capabilities during low-resource fine-tuning, we initialize our policy  $\pi_\theta$  with the weights of the target-language adapted IndexTTS2 model rather than training from scratch. We then employ Group Relative Policy Optimization (GRPO) mixed with a Kullback-Leibler (KL) divergence penalty against the adapted reference model. This ensures that the model explores localized emotional and prosodic improvements while preserving the linguistic adaptation and expressive prior inherited from the original high-resource model.

## 5 Midterm Experimental Results

In this section, we present a toy-example diagnostic experiment designed to illustrate the limitations of the baseline systems, CosyVoice2 and CosyVoice3 (CV2/CV3), on Taiwanese emotional TTS. Rather than serving as a full final evaluation, this experiment uses a controlled synthesis setup to expose the trade-offs between intelligibility, emotion controllability, and speaker fidelity. We compare these baselines with IndexTTS2, our current method in this midterm study, to show that the proposed direction can more effectively handle the observed baseline weaknesses, especially in preserving intelligibility while enabling stronger emotional variation.

### 5.1 Preliminary Baseline Evaluation

As an initial pilot study, we conducted a subjective evaluation to establish a preliminary baseline for Taiwanese TTS. The evaluation used the Mean Opinion Score (MOS) on a scale from 1 to 5 (where 5 represents the best performance). The evaluation panel consisted of 12 members from the SARC laboratory. The test corpus involved 20 sentences for evaluating speech intelligibility and naturalness, and an additional 10 sentences specifically for evaluating voice similarity. In total, the subjective evaluation yielded 600 rating samples ((20 \* 2 + 10) test items × 12 evaluators).

Table 1 shows the preliminary subjective evaluation results of the baseline method, CosyVoice2 [Du et al., 2024], on Taiwanese TTS. The model achieves a similarity MOS of 4.16, an intelligibility MOS of 4.41, and a naturalness MOS of 4.37, resulting in an overall mean MOS of 4.31. These results suggest that a strong off-the-shelf baseline can already generate high-quality Taiwanese speech. This observation motivates our project to move beyond basic intelligibility and naturalness, and to focus on whether emotional expressiveness can be preserved and recovered during low-resource adaptation.

Table 1: Preliminary Subjective Evaluation (MOS) on Taiwanese TTS.

| Model      | Similarity MOS $\uparrow$ | Intelligibility MOS $\uparrow$ | Naturalness MOS $\uparrow$ | Mean $\uparrow$ |
|------------|---------------------------|--------------------------------|----------------------------|-----------------|
| CosyVoice2 | 4.16                      | 4.41                           | 4.37                       | 4.31            |

## 5.2 Evaluation Synthesis Protocol

To compare emotional expressiveness across models on Taiwanese, we synthesized a large-scale evaluation corpus from the TAT-Vol1 texts (2,752 utterances). All systems were conditioned on the same fixed reference speaker audio (a single Taiwanese speaker) to control for speaker identity; only the emotion label was varied across the seven target categories: *angry*, *happy*, *fearful*, *disgusted*, *sad*, *surprised*, and *neutral*. Emotion labels were assigned in a balanced, shuffled order across utterances (random seed fixed at 42 for reproducibility). Table 2 summarizes the inference setup and generation statistics for each system.

Table 2: Evaluation synthesis statistics for the three TTS models on 2,752 TAT-Vol1 utterances.

| Model        | Emotion Control | Audio Duration  | Failures  |
|--------------|-----------------|-----------------|-----------|
| CosyVoice2   | Text prompt     | 07:53:41        | 39        |
| CosyVoice3   | Text prompt     | 09:48:08        | 5         |
| IndexTTS2    | 8-dim vector    | 09:21:57        | 0         |
| <b>Total</b> |                 | <b>27:04:06</b> | <b>44</b> |

## 5.3 Speech Emotion Recognition (SER)

To evaluate whether the intended emotion is recognizable in the synthesized speech, we apply the emotion2vec speech emotion recognition (SER) model to the generated utterances and report the average recognition score for each target emotion. Results are summarized in Table 3.

Table 3: Speech Emotion Recognition (SER) scores broken down by target emotion.

| Model      | Mean | Neutral | Angry | Happy | Sad  | Surprise | Fear | Disgust |
|------------|------|---------|-------|-------|------|----------|------|---------|
| CosyVoice2 | 0.14 | 0.96    | 0.00  | 0.02  | 0.02 | 0.00     | 0.00 | 0.01    |
| CosyVoice3 | 0.13 | 0.67    | 0.00  | 0.05  | 0.14 | 0.02     | 0.00 | 0.03    |
| IndexTTS2  | 0.22 | 0.98    | 0.13  | 0.27  | 0.01 | 0.15     | 0.00 | 0.01    |

## 5.4 Word Error Rate (WER)

We transcribe each synthesized utterance using a locally hosted faster-whisper-taigi-pinyin-large-v7 model, a Whisper-large checkpoint fine-tuned for Taiwanese pinyin transcription. Both the reference (TAT-Vol1) text and the ASR hypothesis are normalized before scoring: hyphens and punctuation (-, ., !?) are stripped, whitespace is collapsed, and both strings are lowercased. For analysis and debugging, the output CSV keeps both the original reference/transcription strings and their normalized forms; all WER scores are computed using the normalized label\_clean and transcription\_clean fields. Word-level edit-distance statistics (hits, substitutions, deletions, insertions) are then computed via the jiwer library. We report average per-utterance WER across the full evaluation set and broken down by target emotion. Failed-generation cases (*Missing*) are excluded. Because Taiwanese romanization ASR remains imperfect, the absolute WER values should be interpreted as a diagnostic proxy rather than as fully reliable intelligibility scores. Nevertheless,

all systems are evaluated with the same ASR and normalization pipeline, so the relative comparison is still informative. Results are summarized in Table 4.

Table 4: Word Error Rate (WER) across the full synthesized corpus and broken down by target emotion.

| Model      | Mean   | Neu.   | Ang.   | Hap.   | Sad    | Sur.   | Fea.   | Dis.   |
|------------|--------|--------|--------|--------|--------|--------|--------|--------|
| CosyVoice2 | 1.6663 | 1.6392 | 1.6884 | 1.6565 | 1.6570 | 1.6879 | 1.6483 | 1.6865 |
| CosyVoice3 | 2.1102 | 2.1140 | 2.1374 | 2.0907 | 2.0798 | 2.1003 | 2.1413 | 2.1077 |
| IndexTTS2  | 1.5089 | 1.5756 | 1.5189 | 1.4641 | 1.5187 | 1.5745 | 1.4617 | 1.4488 |

## 5.5 Speaker Similarity (SS)

Speaker embeddings are extracted using SpeechBrain’s ECAPA-TDNN encoder pre-trained on Vox-Celeb (speechbrain/spkrec-ecapa-voxceleb). Each audio file is mono-mixed and resampled to 16 kHz before being encoded. Speaker Similarity (SS) is then defined as the cosine similarity between the embedding of the synthesized audio and that of a fixed reference recording (a single Taiwanese speaker), so that SS measures how well each model preserves the target timbre across all synthesized emotional variants. Because emotional rendering perturbs the acoustic features used by the speaker encoder, we report both the corpus-wide mean and the emotion-wise breakdown in Table 5.

Table 5: Speaker Similarity (SS) across the full synthesized corpus and broken down by target emotion.

| Model      | Eval. | Mean   | Neu.   | Ang.   | Hap.   | Sad    | Sur.   | Fea.   | Dis.   |
|------------|-------|--------|--------|--------|--------|--------|--------|--------|--------|
| CosyVoice2 | 2713  | 0.5812 | 0.5877 | 0.5864 | 0.5761 | 0.5903 | 0.5773 | 0.5711 | 0.5793 |
| CosyVoice3 | 2747  | 0.5725 | 0.5699 | 0.5749 | 0.5747 | 0.5718 | 0.5698 | 0.5715 | 0.5751 |
| IndexTTS2  | 2752  | 0.4716 | 0.5616 | 0.3246 | 0.5005 | 0.4860 | 0.4358 | 0.4907 | 0.5023 |

**Conclusion.** This toy-example experiment highlights the main weakness of the CosyVoice baselines: although CosyVoice2 and CosyVoice3 preserve speaker timbre more consistently, they produce higher ASR-based word error rates and weaker emotion recognition scores for most non-neutral emotions. As shown in Table 4, despite the limitations of the Taiwanese romanization ASR system, IndexTTS2 obtains the lowest mean WER (1.5089) under the same evaluation pipeline and remains consistently lower than the CosyVoice baselines across most target emotions. The SER results in Table 3 also show that IndexTTS2 obtains the highest mean emotion recognition score (0.22), suggesting that it better addresses the emotional controllability issue exposed by the CV2/CV3 baselines.

Table 5 further reveals the remaining trade-off. IndexTTS2 obtains the lowest average SS (0.4716), with a particularly large drop for high-arousal emotions such as Anger (0.3246), suggesting that stronger emotional rendering can perturb the acoustic features used by the ECAPA-TDNN speaker encoder. Overall, this small experiment supports the motivation of our approach: compared with the baseline systems, IndexTTS2 more effectively handles intelligibility and emotional expression, while speaker identity preservation remains an important constraint. In the next stage, the base models examined in these toy examples will serve as the foundation for our RL-based realignment experiments, where we will focus on further strengthening emotional controllability and intelligibility.

## 6 Expected Contributions of Each Team Member

- **Hao-Chun Hsieh:** Leads the overall project direction, problem formulation, and methodology design. He is also responsible for developing the base codebase, establishing the theoretical foundations, and drafting the project proposal.
- **Pao-Hua Hsu:** Generates the required audio files and refines the project proposal. Also supporting system architecture diagrams and designing the final poster.
- **Nien-Tse Wu:** Extends the base codebase to ensure a lightweight and highly efficient implementation. He also refines the project proposal alongside spearheading empirical evaluation, dataset curation, and hyperparameter tuning.



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